boundaries of modern aerospace engineering. The following key characteristics outline the defining features of fifth-generation drones:

- *Cognitive artificial intelligence (AI) and self-learning systems*
  Fifth-generation drones employ cognitive AI architectures that allow them to think, learn, and evolve based on situational data. Through deep neural networks and machine learning algorithms, these drones continuously analyze their environment, adapt to new challenges, and optimize flight behavior without human intervention. This cognitive capability enables autonomous decision-making, predictive analysis, and mission reconfiguration in real time.
- *Full autonomy and human–machine teaming (HMT)*
  Unlike earlier systems requiring constant operator input, fifth-generation UAVs function with complete operational autonomy. However, they are also designed for collaborative teaming with humans and other machines, an advanced form of human-in-the-loop control, ensuring safety, accountability, and mission adaptability. This balance allows the drone to act independently while maintaining a strategic link with human command structures.
- *Quantum communication and cybersecurity integration*
  These drones incorporate quantum-encrypted communication networks, enabling ultra-secure, instantaneous data exchange between multiple platforms. Quantum key distribution (QKD) ensures resilience against cyberattacks, GPS spoofing, or data interception, providing an unprecedented level of communication integrity and mission confidentiality.
- *Swarm Intelligence 2.0 and collective coordination*
  Fifth-generation drones operate as part of intelligent, self-organizing swarms, leveraging AI-based collaborative behavior. Each unit acts as